



The Yatir Forest from space. The dark color of the forest contrasts with the surrounding, desertified landscapes. The Yatir, which covers about 30 km², warms its local environment by absorbing incoming solar radiation, whereas the surrounding bright desert landscapes reflect more of the incoming radiation to space. Today, the Yatir shows up as a green anomaly in a vast desert landscape, but in biblical times, this entire region was forested.

report a real-world analog to this computer-enabled thought experiment.

Desertification exposes the bright soil surface, which reflects sunlight, much as in the boreal simulation. At the same time, increased convection over dryland forests such as the Yatir cools the surface, reducing outgoing thermal radiation but increasing the radiative forcing on the overlying regional

mate system have interacted over millennia.

More than 3000 years ago, at the dawn of the human modification of the Earth system, the Israelites entered Canaan (modern-day Lebanon, Israel, and the Palestinian territories) and were commanded by Joshua to “go up into the forest country and clear an area for yourselves there” [Joshua 17:15; see also (5)]. Those early settlers released carbon as they cleared forests, thereby changing the albedo, affecting the surface energy balance, and altering the local climate. The modern-day Yatir Forest was planted in 1964 by the Jewish National Fund and—as Rotenberg and Yakir document—has substantially modified the local climate. A global program of dryland reforestation may initially cause regional warming as these new forests modify the surface energy balance, but will pay dividends in the long term as these forests become substantial global carbon sinks.

ronments to expand the scale over which processes are observed.

The authors extend their analysis to consider other pathways through which forests influence climate and to elucidate the role of drylands in the overall surface energy budget of Earth’s land surface. Since Bonan *et al.*’s seminal 1992 study (3), it has been known that tree cover affects the local radiation balance, with important consequences for climate. Forests are dark and absorb incoming solar radiation, converting it into energy for photosynthesis and heat and thereby causing local warming. Bonan *et al.*’s computer simulation explored eliminating the dark boreal forest cover and thereby exposing the bright, highly reflective snow. The bright surface reflects more of the incoming sunlight, cooling the surface. Rotenberg and Yakir now

atmosphere and likely increasing air temperatures aloft. Whereas desert surfaces are hotter than vegetated ones, the atmosphere overlying the desert cools with altitude more quickly and is cooler overall. Paradoxically, desertification has thus likely contributed local cooling to offset the global warming from the carbon release that occurs when dryland forests are cleared.

Although modern humanity has a hard time realizing it, the climate system has never been unchanging. Human activities have long been a driver of change in the Earth system and will continue to be for the foreseeable future (4). Rotenberg and Yakir’s study of the arid Yatir Forest shows how recent desertification has affected local temperatures and global climate. It also provides a perspective on how humans and the cli-

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SYSTEMS BIOLOGY

Amoeba-Inspired Network Design

Wolfgang Marwan

The ability to self-optimize is one of the fundamental properties of living organisms. Adaptive self-optimization in the course of biological evolution is an obvious phenomenon, although it occurs on a time scale of millions of years. On page 439 of this issue, Tero *et al.* (1) describe a model system where self-optimization of cell morphology in response to a chosen experimental situation can be directly observed and mathematically

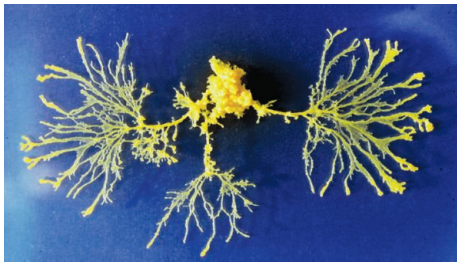
quantified as it occurs on a time scale of hours to a few days. These living cells display adaptive behavior of the sort that may be desired for scalable, multicomponent networks that are supposed to function robustly in the absence of central control mechanisms. Self-organization, self-optimization, and self-repair as it naturally occurs in the slime mold *Physarum polycephalum* are capabilities that may be required for technological systems such as mobile communication networks or networks of dynamically connected computational devices.

The *Physarum* microorganism (a plasmodium) used by Tero *et al.* is a multinucleate

Computational models of biological self-organizing systems may have real-world applications for optimizing dynamic technological networks.

single cell that continually grows as long as nutrition is available. During growth, multiple nuclear divisions occur but the growing cell does not divide, so that all nuclei remain suspended in a single contiguous cytoplasmic volume. When the plasmodium grows on a nutrient-rich substratum, it covers the surface as a coherent layer (like a pancake). If nutrition becomes limited, it forms fenestrae and finally transforms into a network of interconnected veins that enclose the entire cytoplasmic volume (see the figure). Each vein is a gel-like tube covered by a cell membrane and contains a core of fluid cytoplasm.

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Adaptive network formation. The slime mold *Physarum polycephalum* is a single-cell organism that forms a veined network that explores the available space and connects food sources (left). Understanding the formation and behavior of this type of biological adaptive network may inform the development of real-world dynamic technological networks such as mobile communication networks or transportation systems, such as the Tokyo rail system (right).

By rhythmic contraction of its cytoskeleton, cytoplasm is continually pumped through these veins, and this continuous mixing seems to be the reason why all nuclei proceed synchronously through the cell division cycle. The network architecture is highly dynamic. Veins change in thickness, they may form and vanish again, and the plasmodium as a whole can crawl over its substratum, moving over centimeters in a couple of hours. Plasmodia usually do not dissociate. If food sources are spatially separated, such as oat flakes scattered over a wet surface, the plasmodial veins attempt to connect these food sources along the shortest possible pathways, even finding optimized paths through a maze (2). How this optimization is performed in terms of molecular mechanisms remains a challenging question.

Tero *et al.* present a surprisingly simple mathematical model for the development of a network of veins connecting multiple purposefully placed food sources. In their experiment, cities around Tokyo were represented as oat flakes on a wet surface that was inoculated with *Physarum*. Plasmodial veins connected these oat flakes by forming an optimized network closely approaching the purposefully designed Tokyo railway system.

Networks of plasmodial veins form without any central control mechanism that might instruct the organism about the relative position of the oat flakes or tell it how to connect them. The self-optimization algorithm that drives this morphogenesis works equally efficiently for networks with few or many nodes (oat flakes). The absence of central control mechanisms inspired Tero *et al.* to design a corresponding computational model capable of adaptive network design. Using a simple and robust algorithm, the model generates in silico networks that closely resemble those formed by the plasmodium. It is essentially based on feedback loops between the thickness of each tube and the cytoplasmic flow through the respective tube, where streaming

rate positively influences tube diameter and thus its transport capacity for cytoplasm. The dynamics and adaptability of the network structure are the direct results of the iteration of these local rules. The described algorithm or similar ones may provide general solutions for developing real-world, fault-tolerant networks without the need of a centralized control system.

In addition to inspiring next-generation engineering solutions, the work of Tero *et al.* offers a new approach to systems biology. It provides a simple mathematical model for a complex biological phenomenon—adaptive plasmodial shape control—involving growth,

morphogenesis, and homeostasis coordinated with maintenance, self-optimization, and self-repair of morphological and functional structures, the molecular and mechanistic bases of which are completely unclear. Even though the mathematical model has no basis in any specific molecular mechanism, it is nonetheless beautifully useful. It quantitatively mimics phenomena that are fundamental to any living system—phenomena that can be neither captured nor quantified by verbal description alone. Because the model has a number of tunable parameters that can be determined by computational fitting of experimental data, it provides objective readouts (e.g., to search for mutants that display quantitative alterations). Are there genes specifically involved in self-optimization, self-organization, or self-repair? Can molecular circuits be identified, and if so, how are they built, how do they function, and how are they linked to other regulatory systems at the single-cell level? Addressing these questions may lead to yet undiscovered fundamental functionalities in living cells and unravel their molecular basis.

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PALEONTOLOGY

And Then There Were None?

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Direct dating of fossils of the putative last survivors of the Australian megafauna supports claims that they were extinct by 40 thousand years ago and not later.

Giant marsupials, reptiles, and flightless birds once inhabited Australia (see the first figure). But 23 of the 24 genera of these megafauna disappeared in the late Pleistocene (~125 to ~12 thousand years ago). Most Australian megafauna appear to have survived until 51 to 40 thousand years ago, with human impact by hunting or vegetation change proposed as the extinction drivers (1–4). Yet, one site has stood out as an anomaly: Cuddie Springs in interior New South Wales. Persistent claims have been made that this site contains megafauna fossils associated with stone tools in sediments deposited

40 to 30 thousand years ago (5–7), thus indicating prolonged overlap between people and megafauna. These claims have been challenged (2, 8) based on concerns about possible reworking of fossils from older deposits. To resolve this conundrum, Grün *et al.* (9) have now directly dated the fossils themselves. The results provide no evidence for the late survival of megafauna at the site.

Cuddie Springs is an ephemeral lake in the Australian semiarid zone. Fossils were originally discovered when digging wells for water. Many more have since been exhumed from the claypan, including the remains of the giant bird *Genyornis newtoni*, the three-ton *Diprotodon optatum* (the largest marsupial ever to roam Australia), and a range of supersized kangaroos. The fossils found in the same sedimentary layers as stone tools (stratigraphic unit SU6; see the second fig-

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